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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	KALIDERES
Selected Domain	DOMAIN 2
Selected SDG Track (1–17)	SDG 8 and SDG 9
Project / Solution Name	FlowUp
Name (Leader)	Carlos Lee
Email & Phone (Leader)	Carlosleetest2@gmail.com +60 176484186
University Affiliation(s)	Asia Pacific University
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Who: Micro and small grocery retailers in Malaysia.

The problem: Daily operations run across WhatsApp, spreadsheets, paper notes and memory. Sales and stock are recorded in different places, so owners often learn an item is out of stock only when a customer asks for it. Financial figures are hard to verify. Late reorders, expired goods and unchecked records quietly cost margin, and small teams cannot afford complex systems or formal training.

Who is affected: Owners, cashiers, supply staff and finance staff all rely on the same records. Cashiers need reliable product and price information. Supply staff need to catch shortages and check deliveries against purchase orders. Finance staff and owners need figures they can trust before making decisions.

Why it matters: Disconnected records lead to unnecessary purchasing, excess stock and products that sit unsold until expiry. That ties up working capital and creates avoidable waste for businesses with very little margin to lose. The size of these effects must still be measured in participating stores, not assumed.

FlowUp connects sales, inventory and finances in one flow. Employees still check physical stock, review flagged data and approve decisions. The system supports people, it does not replace their judgement.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FlowUp is an integrated, easy-to-use platform that connects sales, inventory and finances for micro and small grocery retailers. It replaces scattered messages, spreadsheets and paper notes with one connected flow.

Automation stock reminder: flags items that need checking or restocking before they run out.

One connected store system: cashier sales update stock automatically. Products, suppliers, finance and budget live in one place, with dashboards and charts for sales, stock and spending.

Role-based control with human approval: owner, cashier, supply, finance and analyst each see only what they need. The system flags data for review, a person verifies or corrects it, and every action is kept in an audit log.

MVP journey: goods received → compared with the purchase order → refund claim approved → stock and supplier score updated → reorder and discount recommendations.

Where AI is used: rules decide the numbers, AI only explains them. Stock reminders, EOQ, holding cost and supplier score come from rule-based calculation logic. The AI receives limited, structured results and writes short explanations and supplier message drafts, with a template fallback if it is unavailable. AI runs through Replit's built-in integration, with no separate AI service. AI output is a suggestion that an employee approves. Missing information is flagged, never invented.

Workshop skills used: GenAI API integration, PostgreSQL database design and cloud deployment on Replit.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Users (owner, cashier, supply, finance, analyst)

↓

FlowUp web app: React + TypeScript (built with Replit Agent)

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Python FastAPI backend: validation + rule-based calculation logic (stock reminder, EOQ, holding cost, supplier score, reconciliation)

↓

PostgreSQL (Replit-hosted, synthetic store data) + audit log

↓

Rule results only (limited, structured data)

↓

AI through the Replit integration → short explanation or supplier message draft (template fallback if unavailable)

↓

Human review and approval

Hosting: Replit Deployments

Sponsor resources: Yes, Replit (sponsor credits plus a self-paid plan). Used for development (Replit Agent), hosting (Replit Deployments), the PostgreSQL database and the AI integration.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Scope first: our first design covered the whole store. We narrowed the MVP to one journey: goods received → compared with the purchase order → refund claim approved → stock and supplier score updated → reorder and discount recommendations. Other modules moved to the roadmap.

Spec before prompting: we wrote a structured specification (roles, data flow, formulas, reference test numbers) before using Replit Agent, then refined the app in the Replit workspace. This gave far better results than short prompts.

Trust by design: every formula is defined once and checked against a reference dataset. Human approval plus an audit log keeps SME owners in control. Rules decide the numbers, AI only explains them.

Deployment: published through Replit Deployments with a PostgreSQL database. It is currently private, so access is arranged for reviewers.

Before Phase 2 ends: compare selected tasks against a manual spreadsheet workflow on equivalent datasets (task time, record accuracy, unresolved discrepancies), add secure individual accounts, test backup and recovery, and run usability sessions with real grocery staff.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 9: Industry, Innovation and Infrastructure** **Target 9.3: support small enterprises' integration into value chains and markets** FlowUp has an indirect alignment with the value-chain integration part of Target 9.3. It connects product availability, supplier records, purchasing and sales information, so small grocery retailers can coordinate their operations and review supplier performance. **SDG 8: Decent Work and Economic Growth** **Target 8.3: support entrepreneurship, innovation and the growth of micro, small and medium-sized enterprises** FlowUp helps small teams spend less time on repetitive record-keeping. This alignment is also indirect. This is limited to operational coordination and information visibility. FlowUp does not provide credit, financial services or direct market access, so it should not be presented as addressing every part of these targets.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Micro and small grocery retailers in Malaysia run sales and stock across WhatsApp, spreadsheets, paper notes and memory. They have to compare these records by hand, which causes delayed updates, duplicate entries and gaps between recorded and physical stock. Owners may also

Metric Category	Your Response
	need extra time to consolidate figures before reviewing performance. **Baseline = the existing workflow before FlowUp.** It is measured through reconciliation time, inventory discrepancy rate, record completeness and the time to prepare a management summary. A verified numerical baseline has not been collected from a retailer pilot yet. We will collect it during an initial observation period with clearly defined tasks and comparable workloads. Synthetic data supports functional testing only. It will not be presented as evidence of real productivity gains.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Core Measurable KPIs** **1. Reconciliation time:** average time to compare inventory and sales records, find differences and finish reconciliation. Measurement: total minutes ÷ reconciliation tasks completed. Improvement (%) = (baseline average − FlowUp average) ÷ baseline average × 100. Supports SDG 8. **2. Inventory discrepancy rate:** share of checked product records whose recorded quantity differs from verified physical stock. Measurement (%) = records with a mismatch ÷ total records checked × 100. A lower rate means better accuracy, and it makes sure speed does not come at the cost of reliable records. **3. Report preparation time:** average time to prepare a complete operational summary from inventory, purchasing and sales records. Measurement: total minutes ÷ equivalent reports completed. Improvement (%) uses the same formula as KPI 1. Supports SDG 8. **4. Cross-record consistency rate:** share of evaluated transactions whose related records are complete and consistent. Measurement (%) = transactions passing all consistency checks ÷ eligible transactions checked × 100.

Metric Category	Your Response
	Example: a completed sale must match its items, quantities, total amount and stock deduction. An accepted goods receipt must match its purchase-order line, accepted quantity and stock addition. Stock must also reconcile: opening stock + goods in − cost of goods sold = closing stock. Supports the value-chain coordination part of SDG 9. **Measurement conditions:** baseline and FlowUp runs use equivalent tasks, workloads and checking procedures. Results from synthetic data are reported as prototype test results, not as proven improvements in real-business performance.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Proposed Phase 2 targets:** **20% reduction** in average reconciliation time compared with the baseline workflow. **20% reduction** in average report preparation time for an equivalent reporting task. **At least 95% cross-record consistency** across the defined evaluation transactions. **No deterioration** in inventory-record accuracy compared with the baseline.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	code	MIT
TypeScript	code	Apache-2.0
Python	code	PSF License
FastAPI	code	MIT
PostgreSQL	code	PostgreSQL License
AI model via the Replit integration	model	Replit and model provider terms

6.3 Employment-Related IP Conflicts

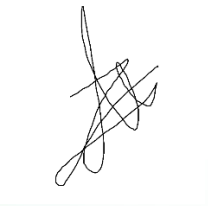
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Carlos Lee]

[25/10/2026]